

ShopEz

Full Stack MERN E-Commerce Application

Project Technical Documentation

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1. Introduction

ShopEz is a comprehensive full-stack e-commerce solution designed to provide a seamless shopping experience. Built using the MERN stack (MongoDB, Express.js, React.js, Node.js), it bridges the gap between consumers and products through a responsive interface.

2. Project Overview

Purpose: To create a robust e-commerce platform allowing users to browse products, manage carts, and complete transactions securely.

Key Features:

- User Module: Auth & Profiles
- Product Module: Search & Filtering
- Cart & Checkout: COD Processing
- Admin Dashboard: Inventory Management

3. System Architecture

Frontend (React): Component-based with Context API state management.

Backend (Node/Express): MVC RESTful API with JWT middleware.

Database (MongoDB): NoSQL storage for Users, Products, and Orders.

4. Setup Instructions

```
git clone https://github.com/thunderfleet/thunderfleet-VIP_C2_ShopEz_E-Commerce_application_by_SIDDANI-SRI-CHARAN.git
```

Install dependencies in both `/server` and `/client` directories using `npm install`.

5. Folder Structure

Directory	Scope
/server/models	Mongoose Data Schemas
/client/src/pages	UI Views (Home, Cart, Orders)

6. Running the Application

Backend: `npm run dev` | Frontend: `npm start`

7. API Documentation

Method	Endpoint	Description
POST	/api/users/login	Auth returning JWT
GET	/api/products	Fetch product catalog

8. Authentication & Authorization

Uses JWT tokens stored in LocalStorage for secure, state-less authentication and role-based access control (RBAC) for admin routes.

9. User Interface

Clean, minimalist aesthetic with responsive navigation and high-resolution media integration.

10. Testing Strategy

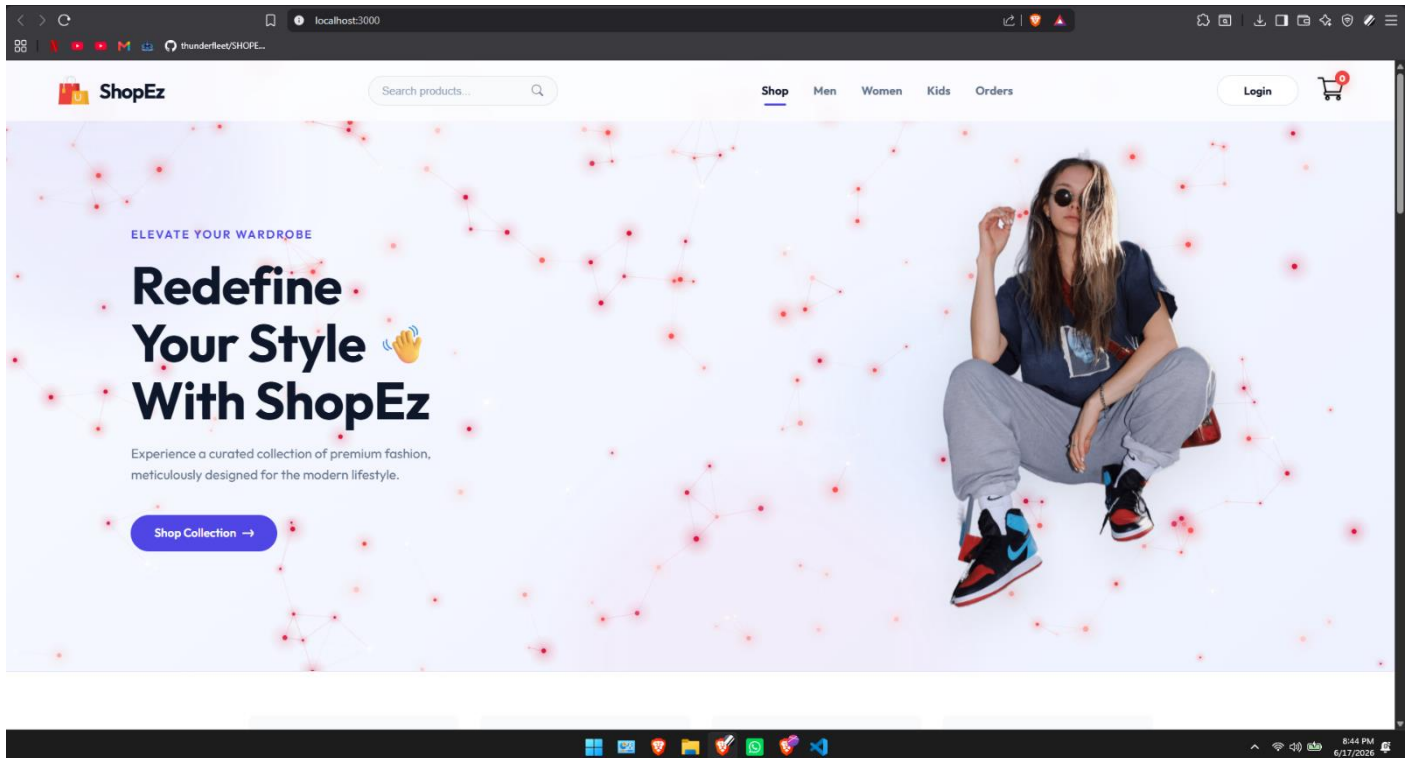
Focuses on Unit testing for logic and Integration testing for database connectivity and API route integrity.

11. Screenshots / Demo

Repository Link: https://github.com/thunderfleet/thunderfleet-VIP_C2_ShopEz_E-Commerce_application_by_SIDDANI-SRI-CHARAN

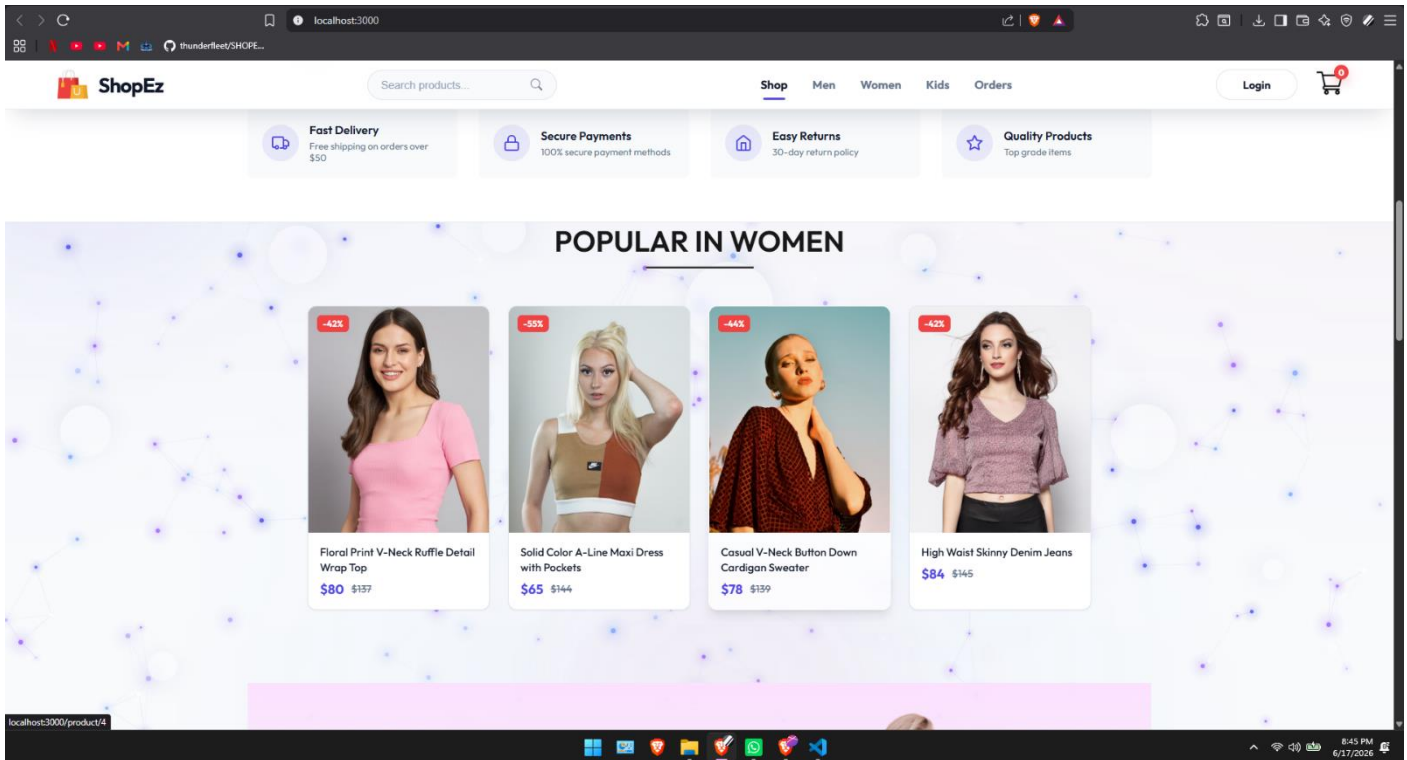
Application Landing Page (Hero Section)

Application Landing Page (Hero Section)



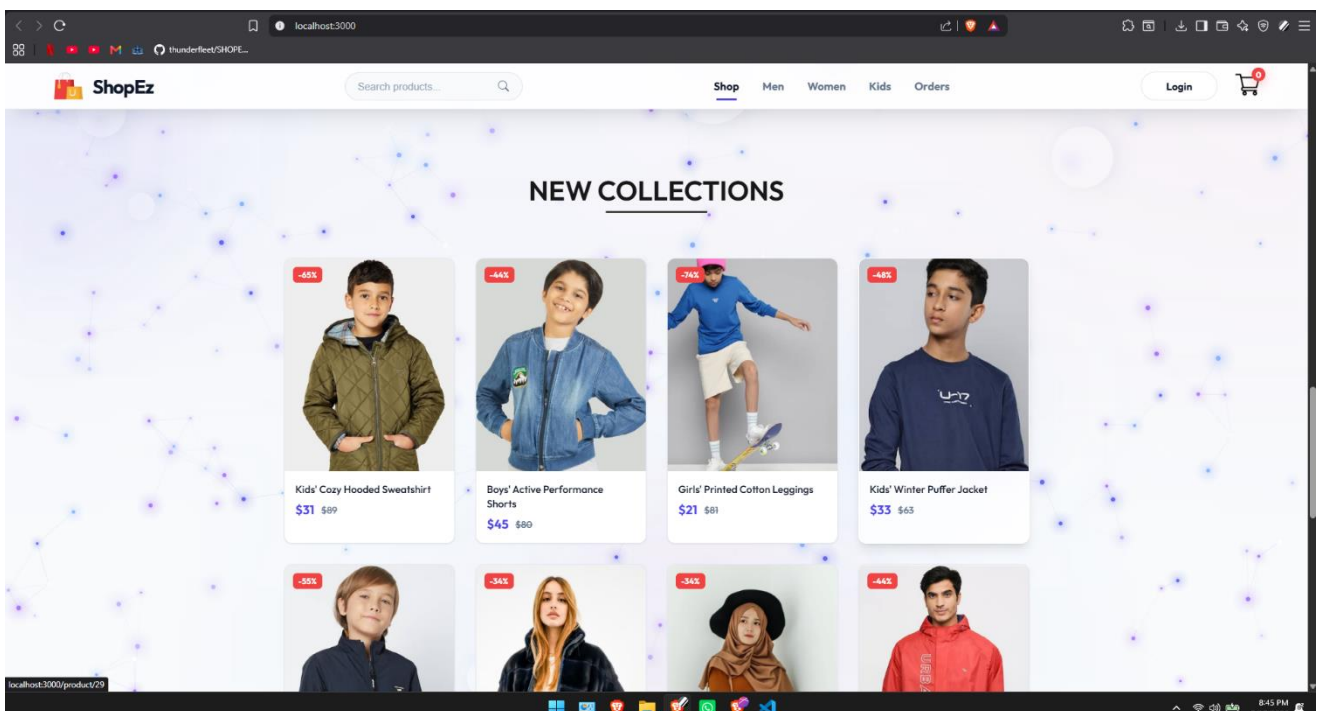
Product Category – Women's Fashion

Product Category - Women's Fashion



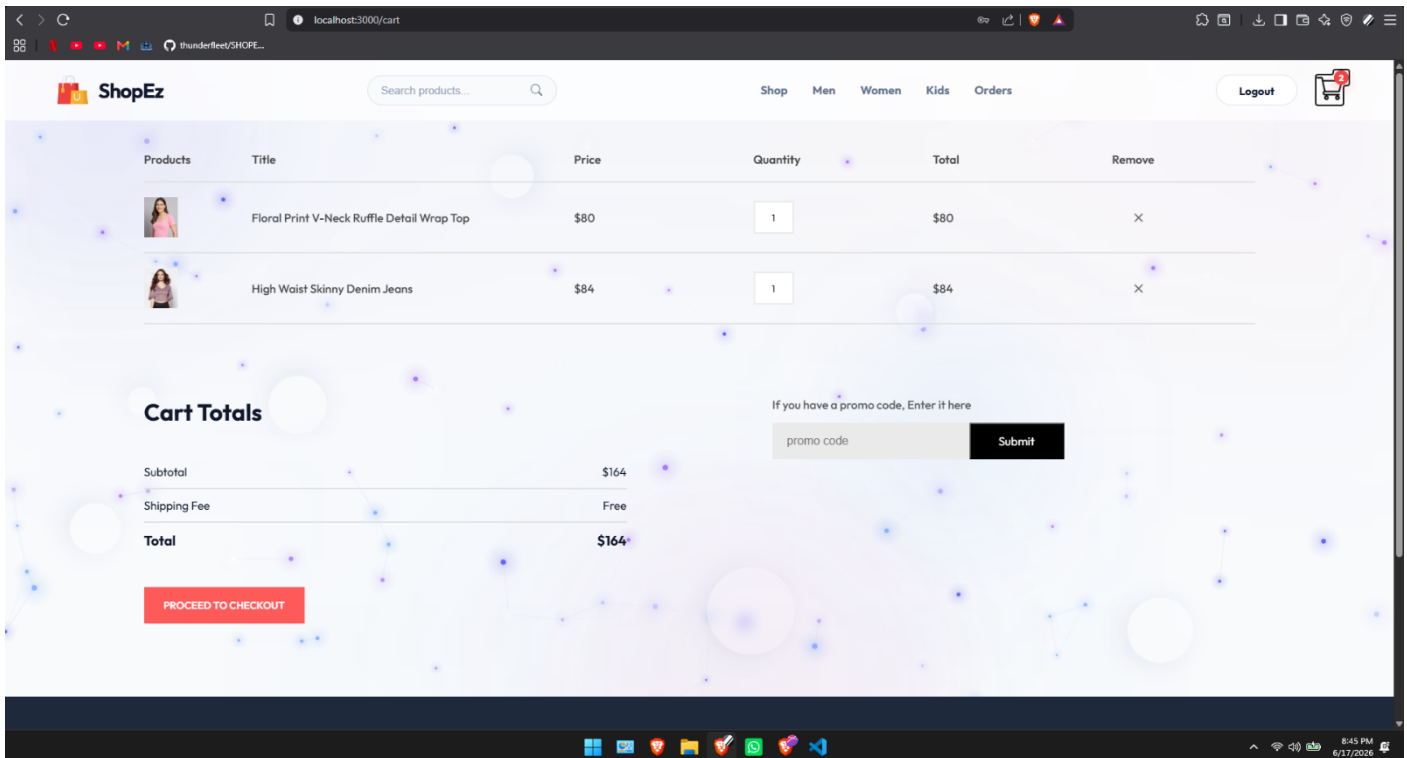
New Collections Section

New Collections Section



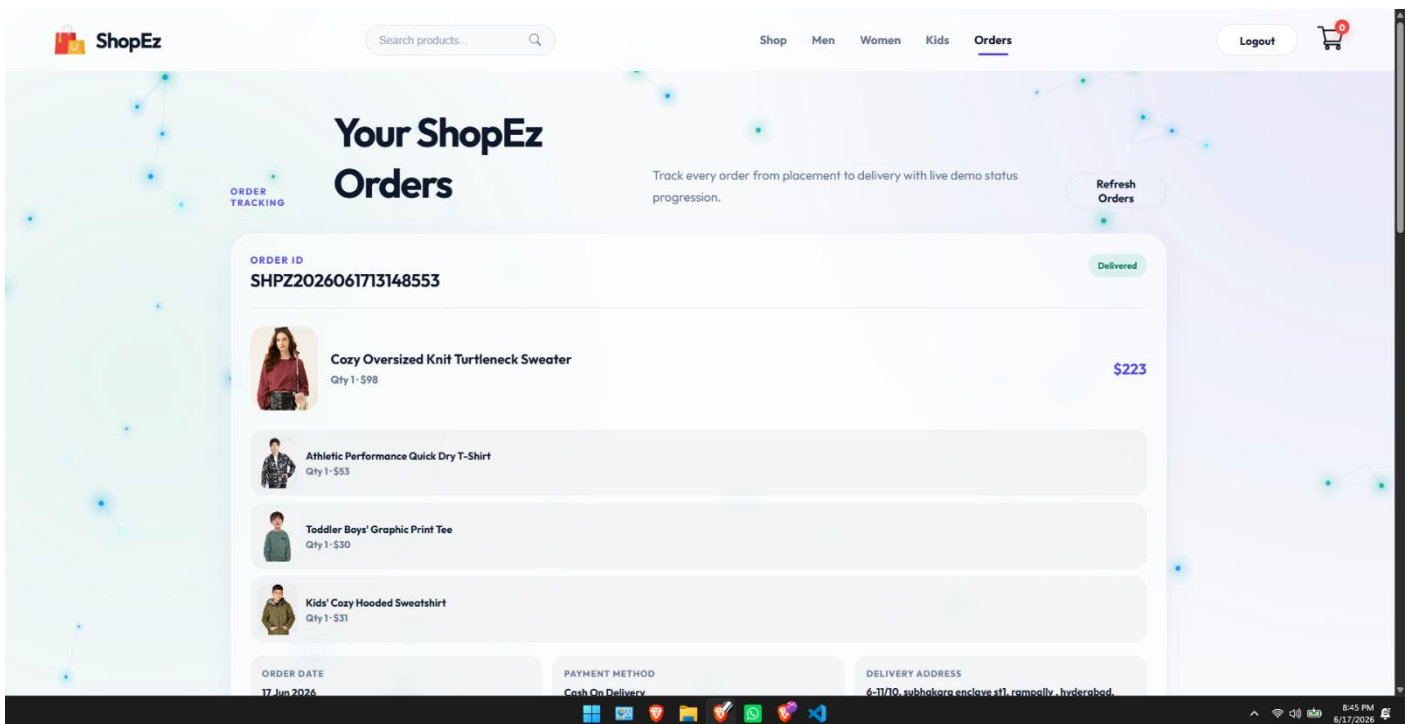
Cart & Checkout Interface

Cart & Checkout Interface



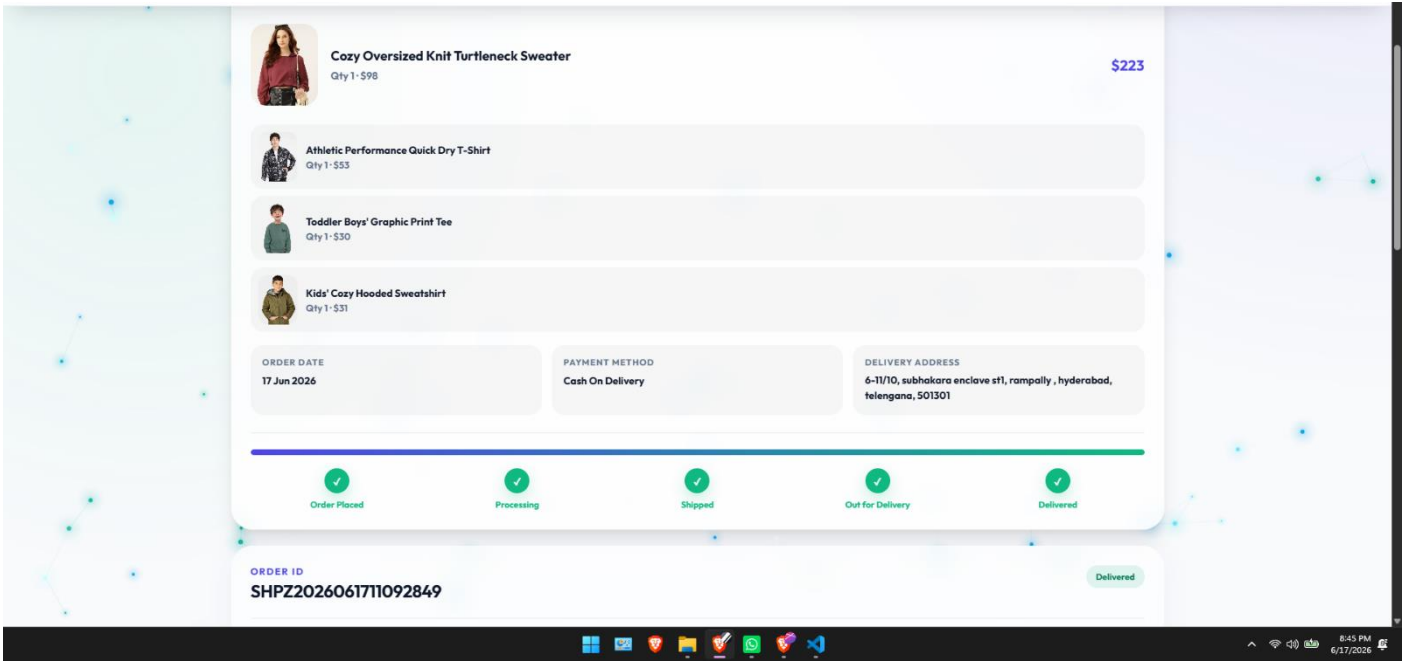
User Orders History

User Orders History



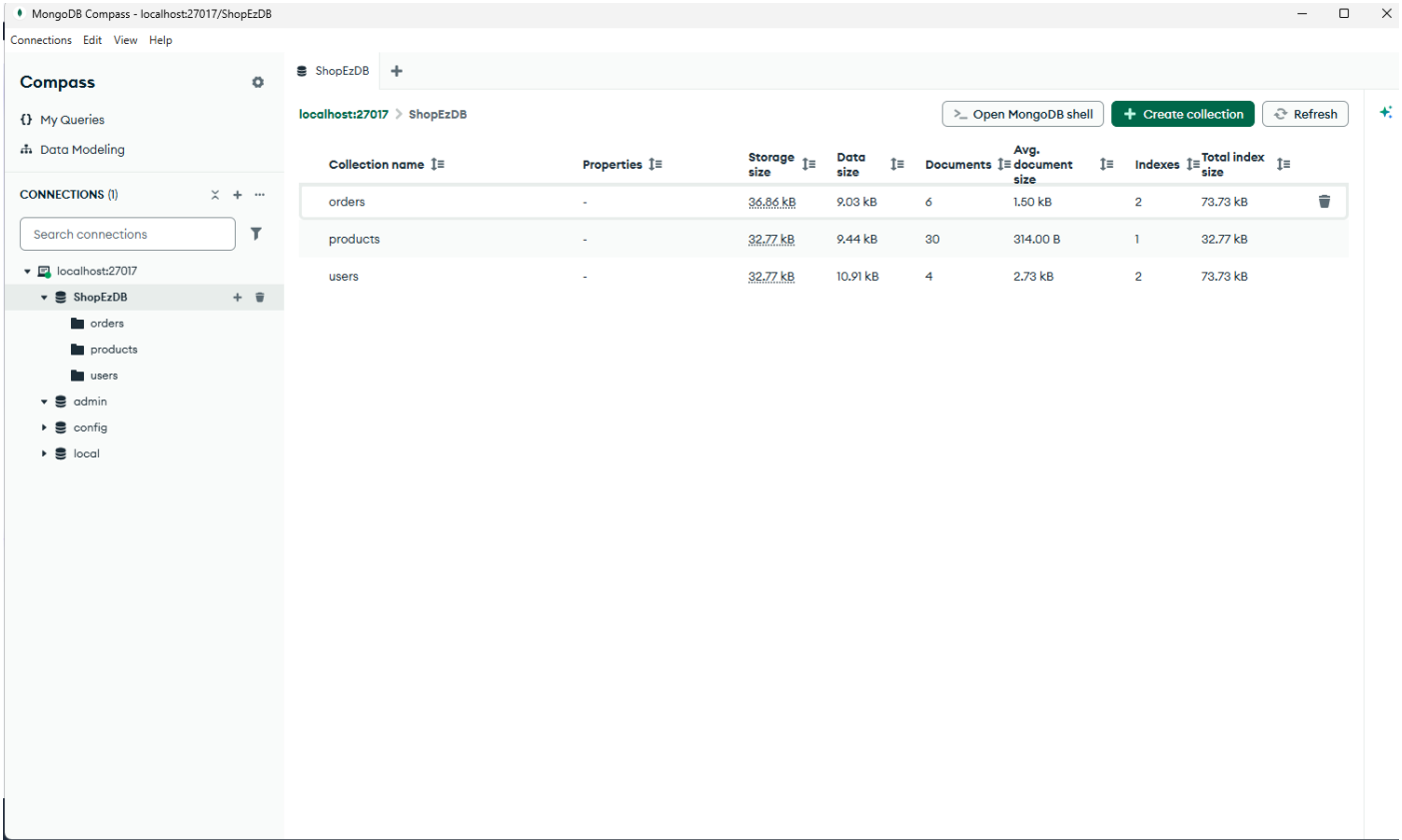
Order Status Tracking System

Order Status Tracking System



Backend Database – MongoDB Collection Overview

Backend Database - MongoDB Collection Overview



Database View – User Credentials & Role Management

Database View - User Credentials & Role Management

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'ShopEzDB' database and its collections: 'orders', 'products', 'users', 'admin', 'config', and 'local'. The 'users' collection is selected. The main panel shows the 'Documents' tab with four user documents. Each document contains the following fields: '_id' (ObjectId), 'name', 'email', 'password', 'cartData' (Object), 'date' (ISODate), and '__v' (0).

```
{ "_id": "6a3009f70fa40360731dc37c", "name": "thunderfleet", "email": "chthunderfleet@gmail.com", "password": "thunderfleet", "cartData": {}, "date": "2026-06-15T14:16:15.797+00:00", "__v": 0 }
```

```
{ "_id": "6a3025de9117601f07c9d869", "name": "lahari", "email": "24eg109a54@anurag.edu.in", "password": "lara45", "cartData": {}, "date": "2026-06-15T16:02:56.629+00:00", "__v": 0 }
```

```
{ "_id": "6a32411b54ef95a51e68868e", "name": "sri", "email": "fleet4735@gmail.com", "password": "thunderfleet", "cartData": {}, "date": "2026-06-17T06:38:23.920+00:00", "__v": 0 }
```

```
{ "_id": "6a324c72e7ab4bdeb69086b8", "name": "charan", "email": "24eg105e51@anurag.edu.in", "password": "thunder", "cartData": {}, "date": "2026-06-17T07:13:50.748+00:00", "__v": 0 }
```

ShopEz Admin

Add Product

Product List

All Products List

PRODUCTS	TITLE	OLD PRICE	NEW PRICE	CATEGORY	REMOVE
	FLORAL PRINT V-NECK RUFFLE DETAIL WRAP TOP	\$137	\$80	WOMEN	×
	SOLID COLOR A-LINE MAXI DRESS WITH POCKETS	\$144	\$65	WOMEN	×
	CASUAL V-NECK BUTTON DOWN CARDIGAN SWEATER	\$139	\$78	WOMEN	×
	HIGH WAIST SKINNY DENIM JEANS	\$145	\$84	WOMEN	×
	SLEEVELESS PLEATED CHIFFON MIDI DRESS	\$131	\$87	WOMEN	×
	WOMEN'S CLASSIC TRENCH COAT	\$102	\$52	WOMEN	×
	ATHLETIC YOGA LEGGINGS WITH POCKETS	\$143	\$71	WOMEN	×

12. Known Issues

- UI re-sync required after certain filter resets.
- Memory optimization for high-res images in server.

13. Future Enhancements

- Integration of Stripe/Razorpay.
- AI Product Recommendations.
- Progressive Web App (PWA) capabilities.